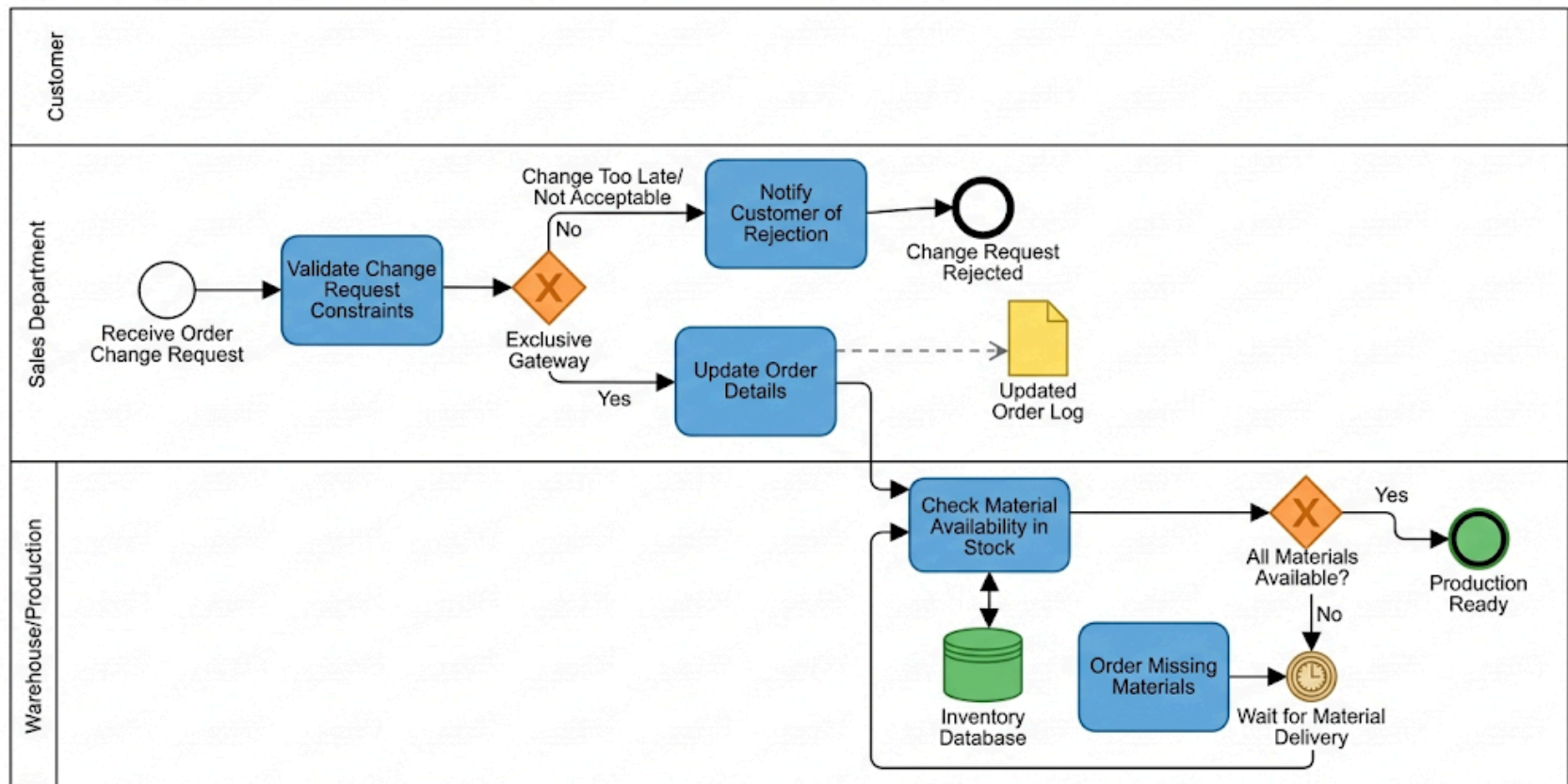


Design and optimize your process diagram

Step 1: Model the process

Build out your diagram here.

BPMN 2.0 — Customer Order Changes and Material Availability Check for RSP Metal Creations



Step 1.2: Model the Process

Process Overview

*This diagram represents the integrated flow of **Order Change Management** and **Material Availability Check** for RSP Metal Creations. It captures the decision points for customer requests and the inventory loops required before production starts.*

BPMN 2.0 Diagram (Conceptual Visualization)

*Aşağıdaki yapı, şablonundaki **"Build out your diagram here"** kısmına yerleştireceğin mantıksal akıştır:*

- **Start Event:** Receive Order Change Request or Initial Order.
- **Exclusive Gateway (Decision):** Is Change Acceptable?
 - **Path A (No):** Notify Customer of Rejection → End.
 - **Path B (Yes):** Update Order Details.
- **Task:** Check Material Availability (Metal and Supplies).
- **Exclusive Gateway (Decision):** Are Materials in Stock?
 - **Path A (Yes):** Proceed to Production.
 - **Path B (No):** Order Missing Materials.
- **Intermediate Timer Event:** Wait for Material Delivery.
- **Loop:** Return to "Check Material Availability" once delivery is received.
- **End Event:** Production Ready.

BPMN Elements & Components Table

Element Type	Name	Purpose in RSP Case
Pool / Lane	Sales & Warehouse	To distinguish between customer-facing staff and production/inventory teams.
Task	Validate Change	Determining if the order can be modified based on time and constraints.
Exclusive Gateway	Decision Point	Handling "Yes/No" outcomes for order changes and stock availability.
Timer Event	Delivery Lag	Representing the waiting period for raw metal shipments.
Data Object	Inventory Record	Connecting the digital or manual stock documents to the checking task.

Step 2: Analyze the diagram

As an analyst, I have reviewed the BPMN 2.0 diagram for RSP Metal Creations and identified the following critical questions and observations to ensure the process is efficient and clear:

- **Clarity of Decisions:**
 - *What specific criteria define the "Acceptable" status for an order change? Is there a clear threshold (e.g., time elapsed or production stage) that the Sales team uses to make this decision consistently?*
 - *In the "All Materials Available?" gateway, is the decision based on real-time digital data or a manual physical count which could lead to errors?*
- **Flow Efficiency & Bottlenecks:**
 - *Does the current communication method for "Notify Customer of Rejection" cause unnecessary delays in freeing up production resources?*
 - *The transition from "Update Order Details" to "Check Material Availability" represents a potential silo; how is information handed over between the Sales and Warehouse departments?*
- **Loops and Wait Times:**
 - *The "Wait for Material Delivery" timer event is a significant bottleneck. How long is the average wait time, and does RSP Metal Creations have a backup supplier to minimize this delay?*
 - *Does the loop back to "Check Material Availability" happen automatically once the shipment arrives, or is there a manual trigger that might cause further idle time?*

Step 3: Optimize and document

Based on the analysis of RSP Metal Creations' current workflows, I propose the following strategic optimizations to enhance efficiency and reduce errors:

1. Optimization for Order Change Management

- **Proposed Change:** Implement a **Standardized Change Policy and Digital Submission Form**.
- **Reasoning:** Currently, the process is "informal," leading to misunderstandings. By establishing clear guidelines (e.g., "Changes only allowed within 24 hours of order placement") and using a digital form, we eliminate back-and-forth communication, speed up decision-making, and minimize the risk of human error.

2. Optimization for Material Availability Process

- **Proposed Change:** Transition from manual checks to a **Digital Inventory Management System with Automated Reordering**.
- **Reasoning:** The current lack of a defined process for material verification causes significant delays. By setting "Safety Stock" levels in a digital system, the software can automatically trigger a purchase order before a shortage occurs. This eliminates the "Wait for Material Delivery" bottleneck and ensures production never stops due to missing raw materials.

3. Loop and Wait Time Reduction

- **Proposed Change:** Integrate an **Automated Notification Loop**.
- **Reasoning:** Instead of manually re-checking stock after a delivery, the system should automatically notify the production team and move the process to the next step immediately upon goods receipt. This reduces "idle time" and improves overall lead time for customer orders.

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Statement of Originality: This BPMN 2.0 model and optimization analysis are my original work. I have designed these structured workflows to solve the informal process challenges of RSP Metal Creations. This study showcases my ability to identify bottlenecks and propose scalable, digital-first solutions aligned with SAP best practices.